Kinds of collisions

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Separate momentum conservation from kinetic-energy conservation.

SUCCESS CRITERIA - I can...

1. Identify the system. State how you handle impulse from outside the system.
2. Recognize sticking as perfectly inelastic.
3. Use kinetic energy to distinguish elastic from inelastic.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “collision” and “elastic” to explain your answer. Sentence starters are optional.

collision: an interaction during which objects exert forces on each other

elastic: a collision that conserves total kinetic energy

inelastic: a collision that transforms some kinetic energy to other forms

01 Warm-up: bring back an earlier idea

A positive triangular force pulse lasts **0.3 s** and peaks at **50 N**. **Find impulse** and velocity change for a **0.6 kg** object.

Word help: impulse = net force acting over time; change in momentum

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $p_{total,\,i} = p_{total,\,f}$; $K = \frac{1}{2}mv^2$.
Negligible external impulse. Elastic collisions also have $K_{total,\,i} = K_{total,\,f}$; sticking transforms some kinetic energy.

Worked reasoning: A **1 kg** cart at **4 m/s** joins a **1 kg** cart at rest. Initial momentum is **4 kg·m/s**. Together, **2 kg** move at **2 m/s**; sticking is one possible collision outcome.

Your turn: Draw before and after snapshots with masses and velocities.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Before **After**

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Sketch before**/after for a collision where two carts stick.
2. **Label one common final velocity**, not two different final velocities.

Sentence starter: Both ... . They differ in ... because ... .

Before contact: add velocity arrows.



Marzano check - circle: ① ② ③

04 Apply the physics

A **2 kg** cart traveling at **+3 m/s** sticks to a stationary **1 kg** cart on a low-friction track.
1. Draw and **label the collision**.
2. **Find the common final velocity**, kinetic energy before and after, and fraction of initial kinetic energy transformed.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Before **After**

Label objects and signed velocities.

Marzano check - circle: ① ② ③

Kinds of collisions

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A classmate says energy is destroyed in an inelastic collision. Rewrite the statement accurately.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A projectile launches horizontally at **6 m/s** from height **45 m**. Use g=**10** and no drag. **Find flight time**, range and final vertical velocity.
Label separate horizontal and vertical motion.

Word help: horizontal = along a level line, usually the x direction

Sentence starter: I found … [unit]. This value means … in this situation.



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3

07 Exit check

Two equal masses collide elastically head-on, one initially at rest. **Give their final velocities relative** to initial moving speed u. Organize the known quantities on the before-and-after diagram.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I notice … . This supports … because … .

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Return to the worked example in problem **2**. For the same initial state, test final velocities (**0**, **4**), (**1**, **3**) and (**2**, **2**) m/s for momentum conservation. Which pairs represent sticking? Organize the known quantities on the before-and-after diagram.

Sentence starter: I notice … . This supports … because … .

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3